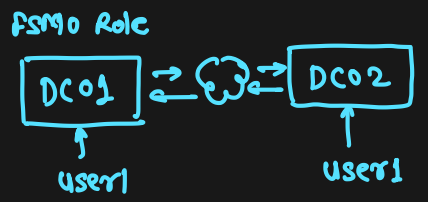
FSMO Roles

Introduction

- FSMO (Flexible Single Master Operations) roles are specialized tasks assigned to specific Domain Controllers (DCs) in Active Directory
- While AD uses a **multi-master replication model** by default, certain operations cannot tolerate conflicts from simultaneous updates across multiple DCs
- FSMO roles solve this by **designating a single DC as the authority** for these specific operations
- **FSMO roles are the exceptions to multi-master replication—operations that must be handled by one DC at a time**
- **Why FSMO roles matter?**
  - **Conflict Prevention**
    - Operations like schema modifications and RID allocation, if performed simultaneously on multiple DCs, would create inconsistencies
    - FSMO enforces single-point authority
  - **Critical Functionality**
    - The PDC Emulator directly affects password authentication and time sync
    - The RID Master affects the ability to create new objects
  - **Security Value**
    - CISA recommends **distributing FSMO roles** across different DCs to increase the difficulty for an attacker to fully compromise the domain—they would need to compromise multiple DCs to seize all roles, giving defenders more detection opportunities



FMSO Roles

- **Forest-Level Roles (One per Forest)**
  - Schema Master
    - Controls all updates and modifications to the Active Directory schema
    - Required when extending the schema (e.g., installing Exchange, running adprep /forestprep)
  - Domain Naming Master
    - Controls the addition and removal of domains and application partitions in the forest
    - Ensures domain name uniqueness
- **Domain-Level Roles (One per Domain)**
  - PDC Emulator
    - Receives priority replication of password changes; processes account lockouts; preferred DC for Group Policy and DFS management; authoritative time source for the domain
  - RID Master
    - Allocates RID (Relative ID) pools to DCs, ensuring new security principals (users, groups, computers) receive unique SIDs
  - Infrastructure Master
    - Maintains cross-domain object references (e.g., when a user from one domain is a member of a group in another domain)
    - Note
      - In environments where all DCs are Global Catalogs, or AD Recycle Bin is enabled, this role has no functional work
      - It should still be assigned to a valid DC to avoid monitoring errors

Pros and Cons



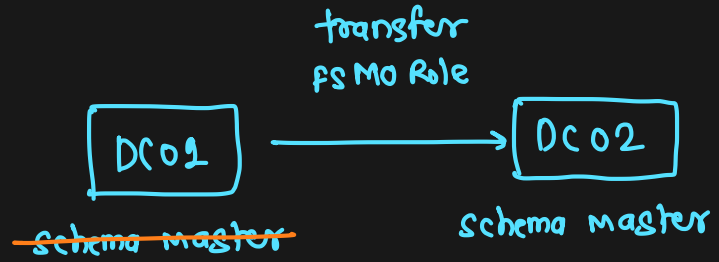
- **Pros**
  - **Conflict Avoidance:** Single-master model ensures uniqueness for critical operations
  - **Clear Accountability:** Administrators know which DC to contact for role-specific operations
  - **Scalability Support:** Enables large distributed environments by isolating operations that don't suit multi-master replication
- **Cons / Risks**
  - **Single Point of Failure:** If a role holder goes offline, role-dependent operations fail
  - **PDC Emulator Load Concentration:** Password changes, time sync, and GPO operations converge on one DC, requiring high-performance hardware
  - **Placement Complexity:** Improper role placement in multi-domain/multi-site environments causes performance issues

Best Practices

- **Role Placement Guidelines**
  - **Configure a standby operations master** for forest-level roles
  - **Configure a standby operations master** for domain-level roles
  - **Place domain-level roles on a high-performance DC**
  - **Leave forest-level roles on a DC in the forest root domain**
  - **In the forest root domain, transfer the three domain-level roles to another high-performance DC**
  - **In all other domains, leave the domain-level roles on the first DC**
- **PDC Emulator Specifics**
  - **Highest workload** of all FSMO roles; requires the most reliable DC
  - The forest root domain's PDC Emulator should synchronize with an **external time source**
  - Ensure 24/7 availability
- **Infrastructure Master Specifics**
  - **Do not** place IM on a Global Catalog server (unless all DCs are GCs)
  - **Exception**: If all DCs are GCs, or AD Recycle Bin is enabled, IM has no functional work, but should still be assigned to a valid DC
- **Security Considerations**
  - **Distribute FSMO roles** across different DCs to increase attacker difficulty
  - Monitor for netdom query fsmo usage, which may indicate attackers enumerating role holders

Transferring FSMO Roles

- Transferring a FSMO role means **moving the responsibility of a specific single-master operation from one Domain Controller (DC) to another DC**
- Example
  - Your company has one **authorized signatory** who can approve payments (this is the FSMO role holder)
  - When that person goes on leave or leaves the company, you **formally hand over** that signing authority to another person
  - After the handover, only the new person can approve payments—the old person no longer has that authority
- In Active Directory terms
  - **One DC** currently holds the role (e.g., PDC Emulator)
  - You **move that role** to **another DC**
  - After the transfer, the **new DC** handles those responsibilities
  - The **old DC** stops performing that role



Transfer process

- **Step 1: Pre-checks**
  - The new role holder (DC2) contacts the current holder (DC1) to request the role
  - The current holder verifies the request is legitimate
- **Step 2: Role Handover**
  - DC1 **releases** the role and records that it no longer holds it
  - DC2 **accepts** the role and records itself as the new holder
  - This change is **written to Active Directory** (specifically to the FSMO-Role-Ownership attribute on the DC objects)
- **Step 3: Replication**
  - The updated role ownership information **replicates** to all other DCs in the domain/forest
  - Until replication completes, some DCs may still think DC1 holds the role (temporary inconsistency)
- **Step 4: DC2 Begins Work**
  - **Important**: DC2 does **not** immediately start performing the role
  - It **waits for inbound replication** of the role's naming context to complete
  - This ensures DC2 has the most up-to-date data before taking over
  - Only after replication completes does DC2 actively perform the role's duties

What Changes After Transfer

Aspect	Before Transfer	After Transfer
Password changes	DC1 (old PDC) receives priority	DC2 (new PDC) receives priority
Time sync source	DC1 serves as authoritative time	DC2 serves as authoritative time
GPO management	DC1 is preferred DC	DC2 is preferred DC
RID pool allocation	DC1 distributes RID pools	DC1 still does this (if RID Master wasn't transferred)
Role ownership in AD	DC1 object holds the role	DC2 object holds the role

Why To Transfer?

- **DC Retirement**
  - You're decommissioning an old DC (e.g., Windows Server 2012 R2)
  - Before shutting it down, you must transfer its FSMO roles to another DC—otherwise, those operations will fail
- **Hardware Upgrade**
  - You're replacing an old physical server with a new, more powerful one
  - Transfer the PDC Emulator (the heaviest role) to the new hardware for better performance
- **Planned Maintenance**
  - You need to take a DC offline for repairs or updates
  - Transfer its roles to another DC first to avoid service disruption
- **Performance Optimization**
  - A DC holding the PDC Emulator is overloaded (slow logons, password sync delays)
  - Transfer the role to a less busy DC.
- **Site/Location Changes**
  - You're restructuring your network
  - Move FSMO roles to DCs in the appropriate sites to reduce replication latency.
- **Security Hardening**
  - Per CISA recommendations, distribute FSMO roles across multiple DCs so an attacker can't compromise the entire domain by taking down one DC

Hands-on

- Check the current FSMO Roles

```
netdom query fsmo
```

- PowerShell Method

```
# Forest-level roles (Schema Master, Domain Naming Master)
Get-ADForest | Select-Object SchemaMaster, DomainNamingMaster

# Domain-level roles (PDC Emulator, RID Master, Infrastructure Master)
Get-ADDomain | Select-Object PDCEmulator, RIDMaster, InfrastructureMaster
```

- Transfer FSMO Roles

```
# Transfer Schema Master
Move-ADDirectoryServerOperationMasterRole -Identity "<TargetDC>" -OperationMasterRole SchemaMaster

# Transfer PDC Emulator
Move-ADDirectoryServerOperationMasterRole -Identity "<TargetDC>" -OperationMasterRole PDCEmulator
```